

Muscle-Gain Nutrition

Foundations of Muscle Gain

Muscle gain is supported by:

1. progressive resistance training;
2. sufficient calorie intake;
3. adequate protein;
4. sufficient carbohydrate to support training;
5. appropriate recovery;
6. adequate sleep;
7. consistency.

Calorie Surplus

A calorie surplus means consuming more energy than the body uses.

A modest surplus is generally more practical than a very large surplus.

A very large surplus does not force all extra energy to become muscle and can lead to unnecessary body-fat gain.

Progress can be monitored using:

- average body-weight trend;
- gym performance;
- strength progression;
- body measurements;
- progress photos when desired.

Protein for Muscle Gain

A commonly useful protein range for healthy exercising adults is approximately:

1.2-2.0 g/kg/day.

Many people doing regular resistance training commonly consume around:

1.6-2.0 g/kg/day.

More protein is not automatically better once intake is already sufficient.

Carbohydrates for Muscle Gain

Carbohydrates can support:

- training intensity;
- training volume;

- glycogen replenishment;
- total calorie intake.

People do not need to avoid carbohydrates while trying to gain muscle.

Useful sources include:

- rice;
- roti;
- oats;
- potatoes;
- fruits;
- whole grains;
- beans;
- lentils.

Dietary Fat for Muscle Gain

Fat provides:

- energy;
- essential fatty acids;
- support for normal hormone function;
- absorption of fat-soluble vitamins.

Useful sources include:

- nuts;
- seeds;
- peanut butter;
- olive oil;
- mustard oil;
- dairy;
- fatty fish.

Protein Timing

Total daily protein intake is more important than consuming protein at one exact time.

It can be practical to spread protein across several meals through the day.

Consuming protein within a few hours before or after resistance training is a reasonable strategy.

Example Vegetarian Muscle-Gain Meal Structure

Breakfast

- oats;
- milk or fortified soy milk;
- banana;
- nuts or peanut butter.

Mid-Meal

- curd or yogurt;
- fruit.

Lunch

- rice or roti;
- dal;
- paneer or tofu;
- vegetables;
- curd.

Pre-Workout

- banana;
- yogurt or another light protein source.

Post-Workout

- a normal protein-containing meal;
- whey or soy protein if convenient.

Dinner

- rice or roti;
- chickpeas, rajma, soy, tofu or paneer;
- vegetables.

Optional Calorie Additions

- nuts;
- seeds;
- peanut butter;
- milk;
- extra whole-grain portions.

Example Non-Vegetarian Muscle-Gain Meal Structure

Breakfast

- oats;

- eggs;
- fruit;
- milk.

Lunch

- rice or roti;
- chicken or fish;
- dal;
- vegetables.

Pre-Workout

- fruit;
- yogurt.

Post-Workout

- protein-containing meal or protein shake.

Dinner

- rice or roti;
- eggs, chicken or fish;
- vegetables;
- curd.

Common Muscle-Gain Misconceptions

More calories always mean more muscle

Excess calories beyond what supports productive training and growth can increase body-fat gain.

Protein supplements are mandatory

Protein powder is convenient but not required if normal food supplies sufficient protein.

Carbohydrates should be avoided

Carbohydrates can support demanding resistance training and recovery.

A supplement can replace progressive training

Muscle growth requires a training stimulus. Supplements cannot replace resistance training.